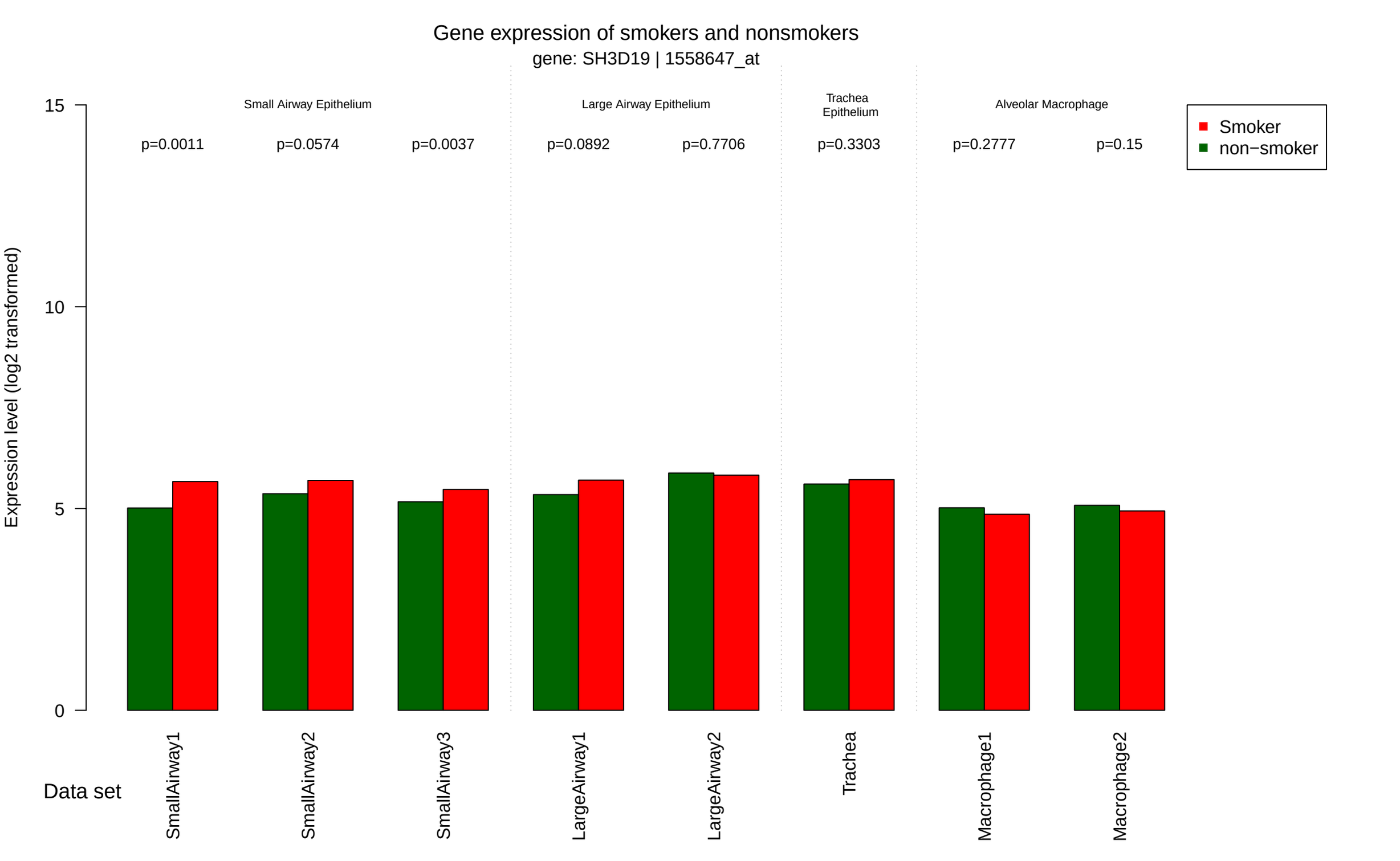
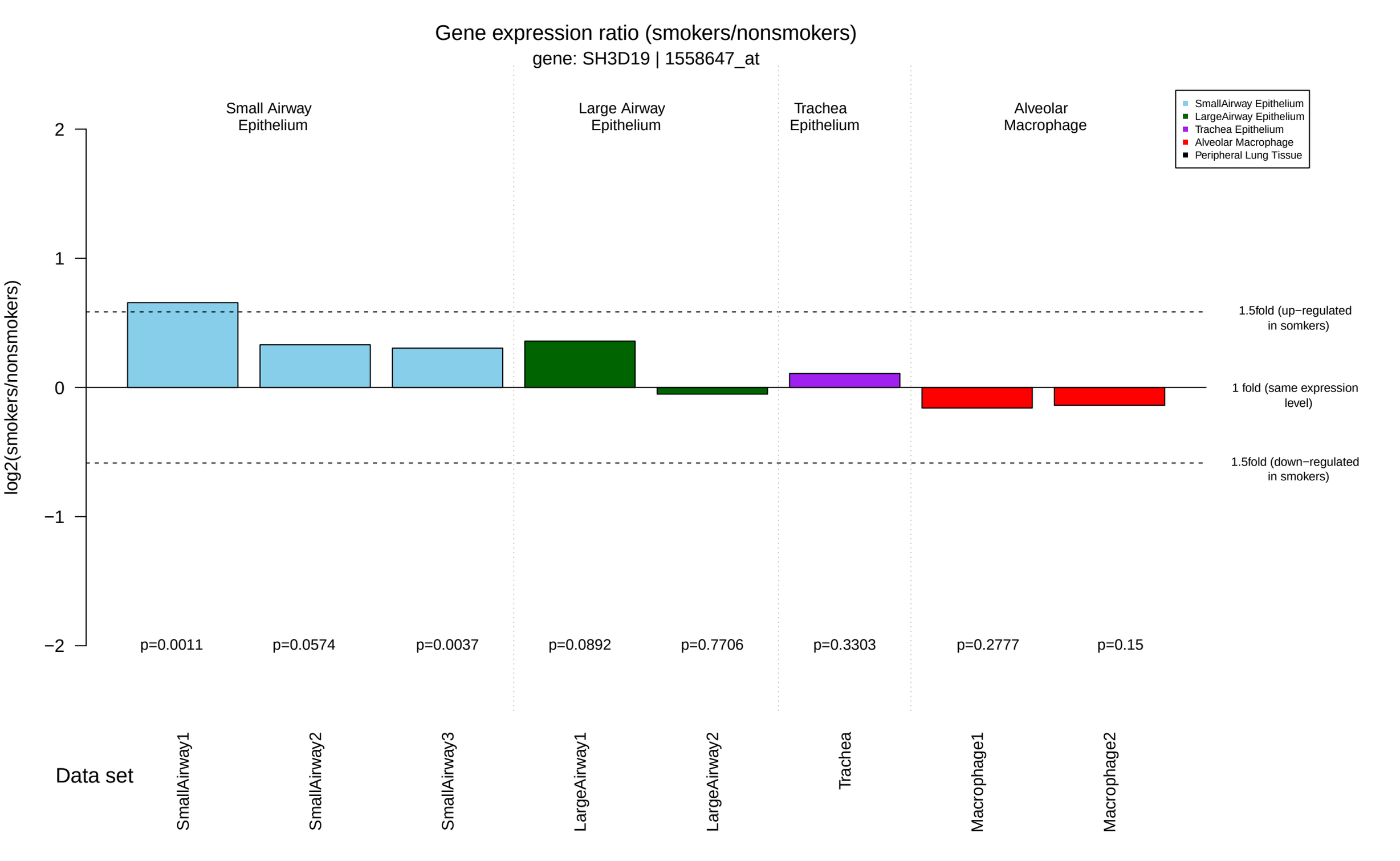
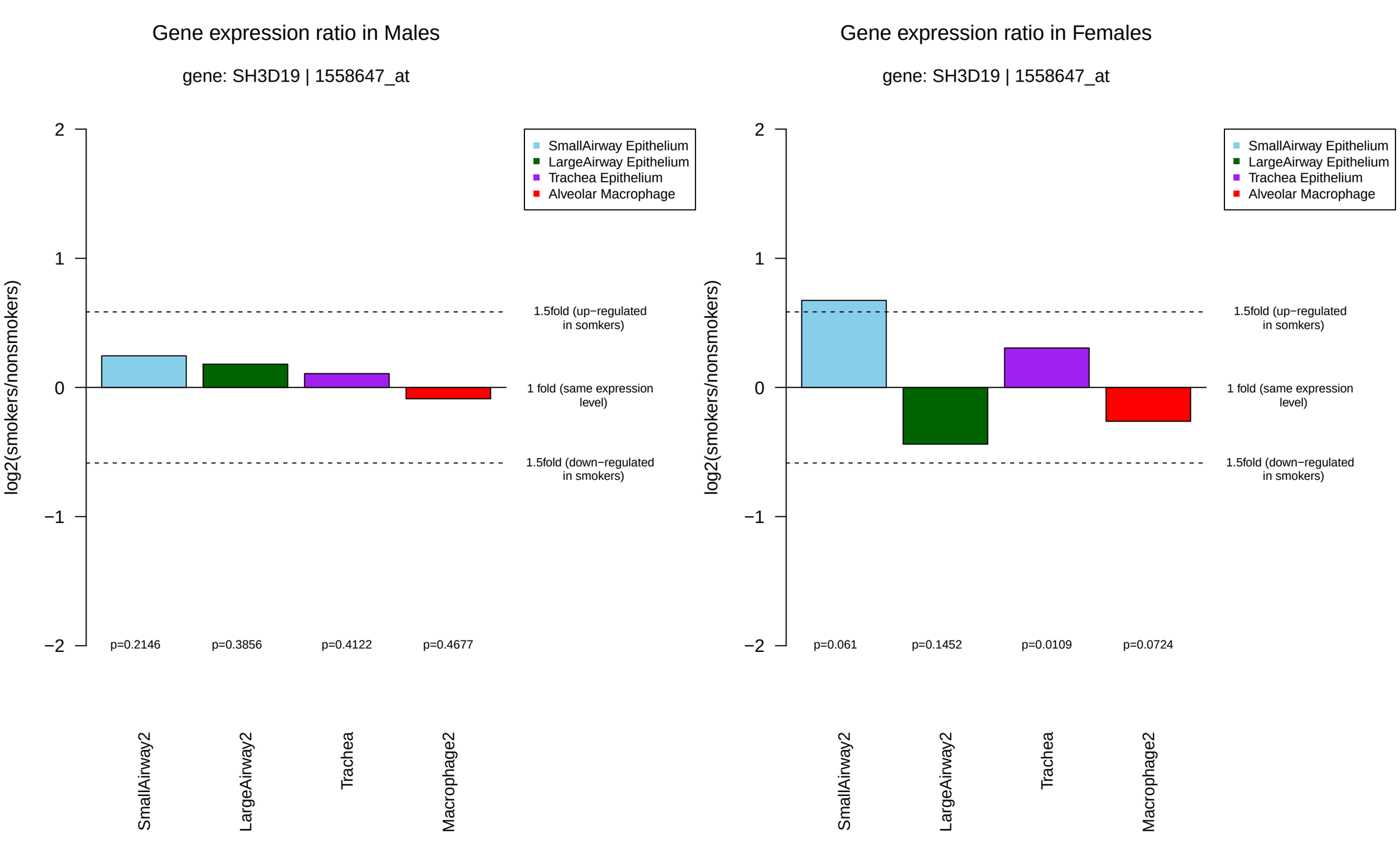
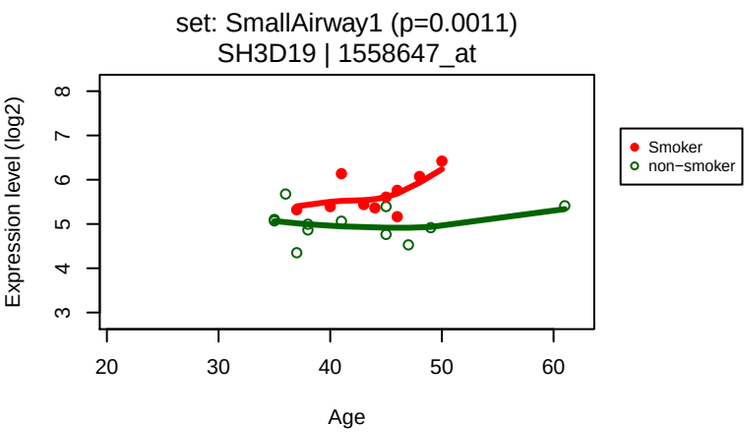
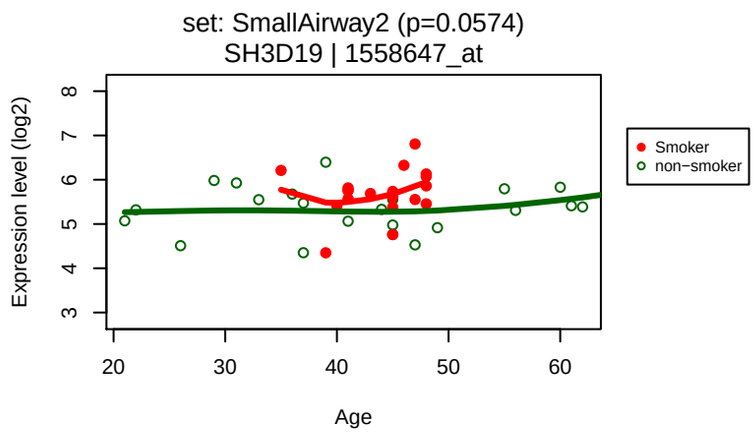
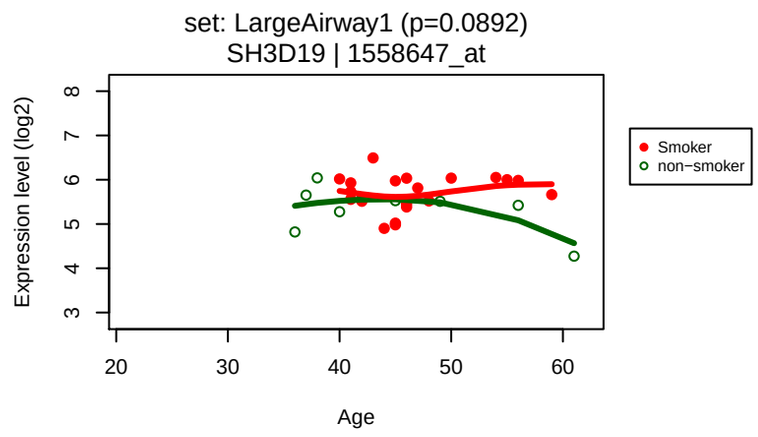

Gene expression of smokers and nonsmokers
gene: SH3D19 | 1558647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0011
p=0.0574
p=0.0037
p=0.0892
p=0.7706
p=0.3303
p=0.2777
p=0.15
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

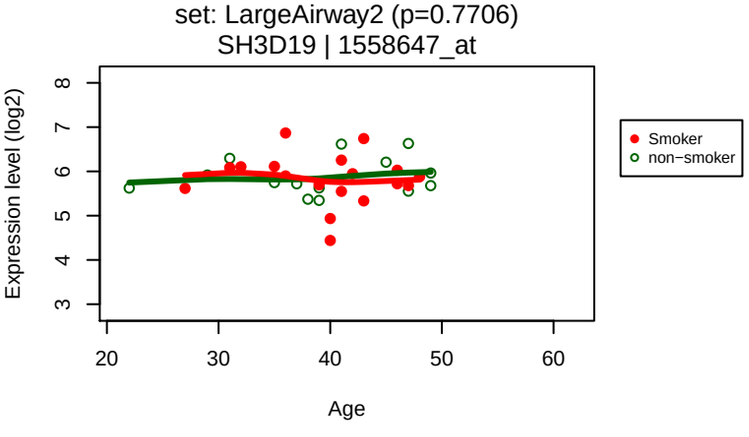

Gene expression ratio (smokers/nonsmokers)
gene: SH3D19 | 1558647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0011
p=0.0574
p=0.0037
p=0.0892
p=0.7706
p=0.3303
p=0.2777
p=0.15
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

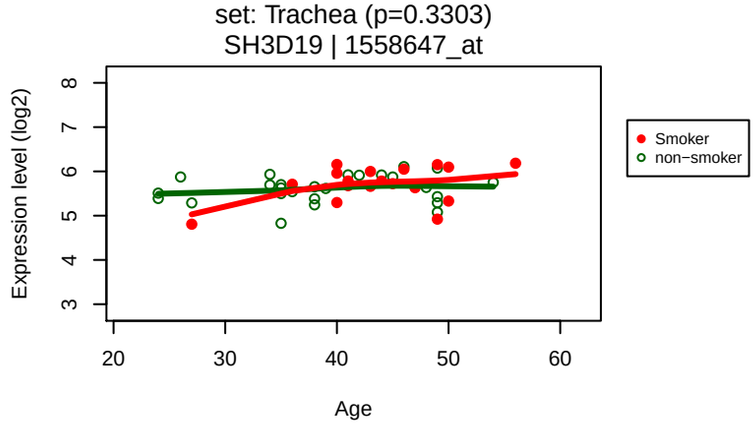

Gene expression ratio in Males
gene: SH3D19 | 1558647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2146
p=0.3856
p=0.4122
p=0.4677
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SH3D19 | 1558647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.061
p=0.1452
p=0.0109
p=0.0724
SmallAirway2
LargeAirway2
Trachea
Macrophage2

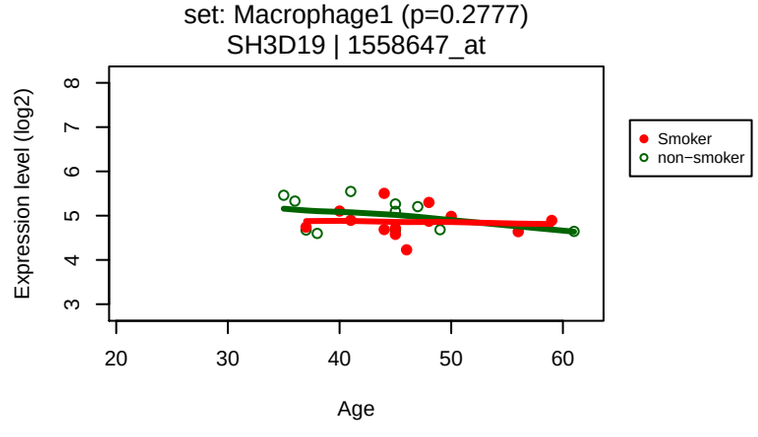

set: SmallAirway1 (p=0.0011)
SH3D19 | 1558647_at
Expression level (log2)
Age
Smoker
non−smoker

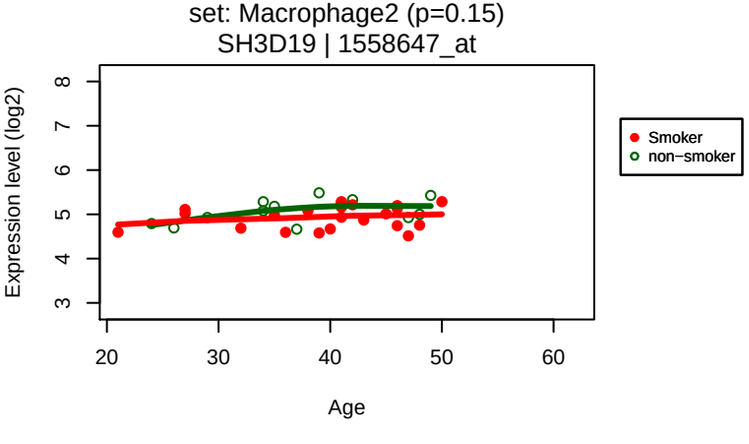

set: SmallAirway2 (p=0.0574)
SH3D19 | 1558647_at
Expression level (log2)
Age
Smoker
non−smoker

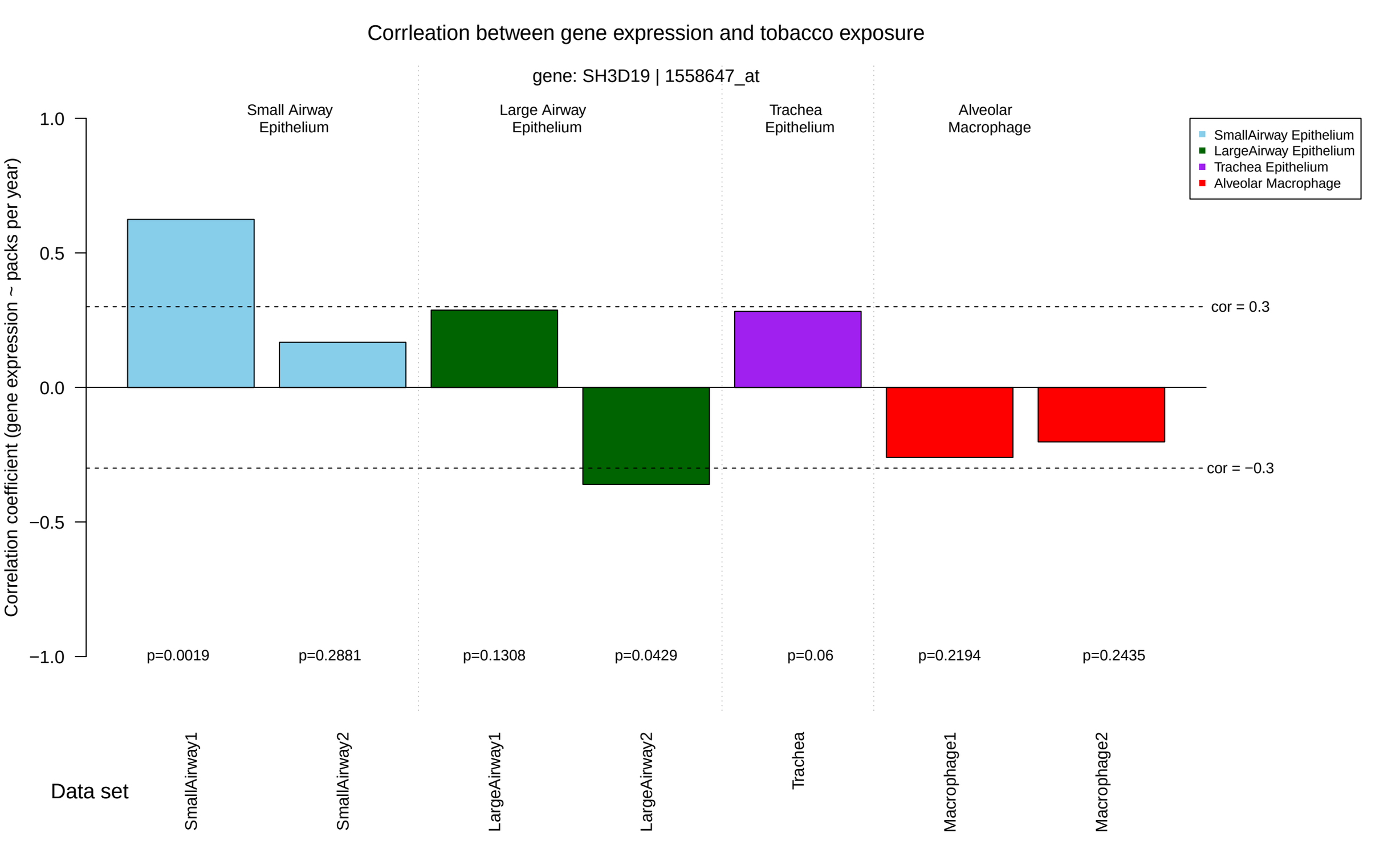

set: LargeAirway1 (p=0.0892)
SH3D19 | 1558647_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7706)
SH3D19 | 1558647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3303)
SH3D19 | 1558647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2777)
SH3D19 | 1558647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.15)
SH3D19 | 1558647_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SH3D19 | 1558647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0019
p=0.2881
p=0.1308
p=0.0429
p=0.06
p=0.2194
p=0.2435
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2